

TASK 2: Hyperliquid Historical Trader Data Analysis

Methodology

- **Execution Profiling:** Processed 211,224 individual trades across 32 accounts. Engineered boolean flags for win/loss, long/short, and aggregated position sizes into standardized buckets (Micro to Whale).
- **PnL & Fee Isolation:** Calculated gross versus net PnL to strictly isolate and quantify the drag of exchange fees on overall profitability.
- **Machine Learning (Clustering):** Applied K-Means Clustering across five behavioral dimensions (Total PnL, Win Rate, Trades/Day, Avg Size, Active Days) to segment the dataset into four distinct trader archetypes: Whales, HFT Bots, Precision Players, and Casual Traders.

Insights

- **The Long Bias Advantage:** In this specific dataset, Long positions significantly outperformed Short positions in both average net PnL and baseline win rate.
- **The Silent Killer (Fees):** Gross profitability was heavily diluted by exchange fees. For high-frequency accounts, trading fees wiped out a massive percentage of gross gains.
- **Archetypes Dictate Outcomes:** A trader's execution style heavily correlated with success. *Whales* rely on massive sizing with moderate win rates, while *Precision Players* rely on high win rates (>75%) with smaller, rapid allocations. *Casual Traders* consistently suffered from overtrading, poor sizing, and fee bleeding.
- **Concentrated Liquidity:** A single coin (HYPE) dominated trading volume, indicating that traders gravitate heavily toward trending narrative assets.

Strategy Recommendations

- **Aggressive Fee Optimization:** Transition from a "Casual" to a "Precision" execution model. Strategies must aggressively utilize limit orders to capture maker rebates rather than paying taker fees. If expected PnL does not cover the spread comfortably, the trade should be skipped.
- **Directional Alignment:** Given the structural outperformance of Longs in this dataset, algorithmic parameters should be weighted to favor Long setups unless strict bearish criteria are met.
- **Capital Allocation:** Avoid high-frequency, low-conviction trades. Mimic the successful clusters by reducing daily trade frequency and increasing position size strictly on high-probability setups in high-liquidity coins.